AYUSHMAN SRIVASTAV

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PROFESSIONAL SUMMARY

B.Tech Aeronautical Engineering student (2023–27) at FGIET Rae Bareli with CGPA 7.8 . Specialized in aerospace design and simulation — completed two end-to-end solo projects: CubeSat LEO Mission (Fusion 360 + ANSYS FEA/Thermal + MATLAB orbital simulation) and Quadcopter UAV (Fusion 360 + ANSYS Structural + ANSYS Fluent CFD + MATLAB Simulink PID). ISRO-certified in Remote Sensing and Space Technology. Experienced in NASA GMAT satellite orbit simulation. Seeking internships in aerospace design, UAV/drone systems, and defence technology.

EDUCATION

B.Tech — Aeronautical Engineering *Sep 2023 – Jul 2027*
Feroze Gandhi Institute of Engineering and Technology (FGIET), Rae Bareli | CGPA: 7.8
(Dr. A.P.J Abdul Kalam Technical University, Lucknow)

Intermediate (Class XII) *Apr 2021 – Feb 2023*
Kamla Nehru Institute of Child Education, Sultanpur, Uttar Pradesh

WORK EXPERIENCE

Engineering Intern — MakeX Program *Aug 2025 – Sep 2025*
Kalam Pragati (ERA Foundation × AKTU) | Lucknow, Uttar Pradesh

- Worked on engineering innovation challenges applying aeronautical concepts to real-world problem statements
- Led hardware integration for Cardio Guard IoT project — designed enclosure in Onshape and fabricated via FDM 3D printing

Space Technology Intern *Aug 2024 – Sep 2024*
Space Technology and Aeronautical Rocketry (STAR) | Surat, Gujarat

- Simulated satellite orbit trajectories using NASA's General Mission Analysis Tool (GMAT) — enhanced LEO mission planning accuracy
- Configured spacecraft parameters, force models, and propagators to optimize Low Earth Orbit trajectories
- Analysed orbital behaviour, ground tracks, and mission performance metrics through detailed simulations

PROJECTS

Quadcopter UAV — Design, CFD & PID Flight Simulation *Mar 2026 – Apr 2026*
Solo Project | FGIET Rae Bareli | Fusion 360 · ANSYS 2026 R1 · ANSYS Fluent · MATLAB Simulink

- Designed complete X-450 quadcopter assembly in Fusion 360 — CFRP hub, arms, motor mounts, landing gear & 10×4.5 twisted aerofoil propellers
- Static Structural FEA in ANSYS under 14N motor thrust — Max Von Mises: 4.296 MPa | CFRP yield: 600 MPa | Safety Factor: ~140×
- ANSYS Fluent CFD on propeller at 6000 RPM (k-omega SST) — Lift Cl: 0.35 | Drag Cd: 0.095 | L/D: 3.68 | Max tip velocity: 41.4 m/s
- PID altitude controller in MATLAB Simulink — stable hover with <10% overshoot

CubeSat Structure Design & Thermal Analysis — LEO Mission *Jan 2026 – Feb 2026*
Solo Project | FGIET Rae Bareli | Fusion 360 · ANSYS Workbench 2025 R2 · MATLAB

- Designed 1U CubeSat (10×10×10 cm) modular assembly in Fusion 360
- Structural FEA under 10g launch load — Max Von Mises: 6.55×10^{-4} MPa | Safety Factor: ~420,000×
- Thermal analysis — Hot case: +120.39°C | Cold case (eclipse): -100°C uniform distribution
- MATLAB orbital simulation — Velocity: 7.62 km/s | Period: 94.47 min | 15.2 orbits/day | 3-orbit ground track

CardioGuard — Real-Time Heart Monitoring & Alert System *Aug 2025 – Sep 2025*
IoT Project | Kalam Pragati Internship | Lucknow | Arduino · ESP8266 · Onshape · ThingSpeak

- Built wearable IoT device using AD8232 ECG + MAX30102 sensors with ThingSpeak dashboard and Telegram emergency alerts
- Designed 3D-printed casing in Onshape — successfully built and tested complete working prototype

CERTIFICATIONS

- From Code to Flight — MATLAB & Simulink | Vyadh Aerospace Pvt. Ltd. | Mar 2026
- Drone Systems & Controls | NPTEL | Nov 2025
- AutoCAD Certified | Autodesk / Udemy | Aug 2025
- Space Tech for Disaster Risk Reduction | ISRO | Jun 2025
- Remote Sensing & Digital Image Analysis | ISRO — IIRS | Mar 2025
- MATLAB Certified | Great Learning | Aug 2024

CORE COMPETENCIES

CAD & Design	Fusion 360, SolidWorks, AutoCAD, Onshape, Autodesk Tinkercad
Simulation	ANSYS Workbench, ANSYS Fluent (CFD), MATLAB Simulink, NASA GMAT
Programming	MATLAB, Python, C, C++, HTML/CSS
IoT & Embedded	Arduino, ESP8266, ThingSpeak, Telegram Bot API
Aerospace	CubeSat Design, FEA, CFD (k-omega SST), PID Control, Orbital Mechanics, Remote Sensing

LANGUAGES

Hindi: Native / Bilingual **English:** Professional Working Proficiency